TIFFANY LIOU

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EDUCATION

M.S. Physics, Northwestern University**,** Evanston, IL — Dec 2025
B.S. Physics, University of California – San Diego, La Jolla, CA — Jun 2024

RESEARCH EXPERIENCE

Graduate Researcher, *Northwestern University* — Sept 2024 – Dec 2025
Advisors: Dr. Xinfeng Xu, Dr. Allison Strom

- Constrained galactic outflow scaling relations using 98 galaxies from the Keck Baryonic Structure Survey and the Keck Lyman Continuum Survey, resulting in the largest outflow sample at Cosmic Noon to date
- Built an automated Python pipeline to perform statistical analysis, fit nebular emission-line profiles, visualize data, and detect outflow signatures

Space Telescope Science Institute Intern, *Society of Physics Students* — June 2023 – Aug 2023
Advisor: Dr. Nimisha Kumari

- Analyzed scaling relations of 98 Lyman-alpha emitter galaxies at $z \sim 2$–4 from the MUSE Hubble Ultra Deep Field, uncovering rest-UV properties and available emission lines for analysis
- Developed Python code that integrates the MUSE Python Data Analysis Framework package to automate data extraction, spectral-fitting, and galaxy candidate selection

Undergraduate Researcher, *University of California, San Diego*
Advisor: Dr. Adam Burgasser — Jul 2024 – Aug 2024

- Reduced spectra of brown dwarfs and asteroids to contribute to the development of the SpeX Archive database
- Implemented Markov chain Monte Carlo (MCMC) analysis to compare JWST/NIRSpec and JWST/MIRI spectra of Y-type brown dwarfs to low-temperature spectral models

Advisor: Dr. Tongyan Lin — Nov 2022 – June 2024

- Developed Python scripts, integrating the Gala package, to simulate stellar streams and dark matter interactions
- Implemented MCMC methods that to successfully infer the mass, size, and 6-D kinematic information of the perturbing dark matter subhalo

CASSI Intern, *Carnegie Observatories* — Jun 2022 – Aug 2022
Advisor: Dr. Josh Simon

- Performed exploratory data analysis and statistical testing to identify patterns in N-body stellar stream simulations
- Created data visualizations using matplotlib and ParaView to understand the impact of dark matter perturbations

PUBLICATIONS

- **Tiffany Liou**, Xinfeng Xu, Allison L. Strom, Nathalie A. Korhonen Cuestas, Claude-André Faucher-Giguerè, Tim B. Miller, Gwen Rudie, Ryan F. Trainor, Naveen A. Reddy, Charles C. Steidel, Yuguang Chen, “Scaling Relations of Galactic Outflows Across Cosmic Time: New Insights from Cosmic Noon” *(submitted to Astrophysical Journal)*
- **Tiffany Liou** & Nimisha Kumari, “Rest-UV Properties of MUSE DR2 Galaxies,” 2024, BAAS, 56, 2
- The SPS Observer: Lyman-Alpha Emitters Offer a Peek into Galactic Evolution , 2024
- **Tiffany Liou** & Josh Simon, “Measuring Dark Matter with Stellar Streams,” 2023, BAAS, 55, 2

AWARDS

Chambliss Award, *243rd American Astronomical Society Meeting* — Jan 2024
Physical Sciences Dean’s Undergraduate Awards for Excellence, *UC San Diego* — Jan 2024

SKILLS

- **Computer Languages:** Python, SQL, MATLAB, LaTeX, Mathematica, HTML
- **Data Analysis:** Code development, Bayesian statistics, machine learning, data cleaning, large datasets

PRESENTATIONS

Galactic Outflow Properties at Cosmic Noon Dec 2025
Chen Group Meeting, *University of Chicago*
Stellar Hallo Group Meeting, *University of Michigan*

Harder, Better, Faster, Stronger – Warm-ionized Outflows at Cosmic Noon Oct 2025
Northwestern Physics and Astronomy Graduate Student Council PAECRS Seminar

Rest-UV Properties of MUSE HUDF DR2 Galaxies
iPoster: 243rd American Astronomical Society Meeting Jan 2024

Measuring Dark Matter with Stellar Streams
Stellar Stream Group, *Vera C. Rubin Observatory* May 2023
iPoster: 241st American Astronomical Society Meeting Jan 2023

TEACHING EXPERIENCE

Teaching Assistant, *Northwestern University* Sept 2024 – Dec 2025
- Facilitated undergraduate physics laboratory sessions and enhanced student learning through group discussions and hands-on experiments
- Provided thorough constructive feedback to students through office hours and assignment grading
- Courses supported: Mechanics (Phys 131–A), Electricity and Magnetism (Phys 131–B), Modern Physics (Phys 131–C)

Supplemental Instruction Leader, *UC San Diego* Jan 2022 – Jun 2024
- Designed and facilitated weekly 80-minute in-person and virtual topic sessions for undergraduate physics students, supporting over 200 unique individuals with a session return rate of 83%
- Fostered active learning environments through implementing pedagogical principles of collaboration, scaffolding, and redirection to promote problem-solving skills
- Courses supported: Mechanics (PHYS 2A, 4A), Electricity and Magnetism (PHYS 2B, 4C), Thermodynamics & Fluids (PHYS 4B), Optics & Modern Physics (PHYS 4C)

Physics Team Course Leader Jan 2022 – Jun 2024
- Bridged communication between the Supplemental Instruction Program Coordinators and the Physics Team through leading weekly team meetings and trainings
- Conducted session observations and ensured the quality of academic support by effectively providing constructive feedback and resources to the session leaders

MENTORSHIP AND LEADERSHIP EXPERIENCE

Letters to a Pre-Scientist, *Remote* Sept 2025 – Present
- Mentored one middle school student through a pen-pal program, discussing topics that include: pathways into STEM, overcoming adversity, and recent breakthoughs in scientific research

STARTastro Graduate Mentor, *University of California, San Diego* Jul 2024 – Aug 2024
- Mentored 8 students from the San Diego area community colleges through the STARTastro summer program to kickstart their academic careers in physics and astronomy
- Designed and led mentorship sessions on imposter syndrome, sense of belonging, and resources for transfer students

CASSI Alumni Mentor, *Carnegie Observatories* Jun 2023 – Present
- Served as a peer mentor for summer interns, providing both scientific and career guidance

President, *Society of Physics Students at UC San Diego* Mar 2023 – Mar 2024
- Led a team of 25 students to promote student interest in physics through event planning, including: scientific outreach, student talks, community town halls, and chats with speakers within physics and astronomy.